

Umair Hussain

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Data Scientist at ChampionX (now part of SLB) focused on developing AI and machine learning solutions for the oil & gas industry. I bring a strong technical foundation in computational mechanics, with deep expertise in the finite element method (FEM), phase field modeling, and physics based simulation from my PhD research at IIT Madras. I specialize in integrating data-driven AI approaches with physics-informed and FEM-based modeling to solve complex real world engineering challenges.

Skills & Relevant Course Work

Technical	📖 Finite element method Computational mechanics Multiphysics modeling Machine learning Parallel computing Phase field modeling Microstructure modeling Plasticity.
Coding	📖 C++, Python, MATLAB, LaTeX.
Softwares	📖 deal.II, Abaqus, TensorFlow, MicroSim, Paraview, SageMath, GMSH & LINUX.
Courses	📖 Applied Finite Element Analysis Advanced Phase Transformations Supervised Machine Learning Variational Principles in Mechanics Engineering Plasticity.

Work Experience

Nov 2024 – April 2025	📖 Research Analyst- Intern , ChampionX (now part of SLB). - Developed a FEM model for the sucker rod pump to predict the pump card. - Predicting broken shaft from sensor data, using Machine Learning .
May 2025 – Present	📖 Data Scientist , ChampionX (now part of SLB). - Working on different AI and physics based models for different artificial lift systems , like sucker rod pumps and ESPs, as well as corrosion prediction project.

Research Experience

Ph.D. Thesis: Electrochemical responses during lithiation of high capacity anode in Li-ion battery

Supervisors: Dr. Narasimhan Swaminathan & Dr. Gandham Phanikumar

- 📖 Developed a **multiphysics** model using **phase field modeling** to investigate the electrochemical response during anode (dis)charging. The model solves **chemical diffusion** under **stress**, coupled with **electrochemical** reactions via Butler-Volmer boundary conditions.
- 📖 Demonstrated the **impact of phase transformation, size of electrode** as well as **stresses** on electrochemical responses via variations in voltammogram peaks. Highlighted the **effects of stresses** on **phase transformation** using voltammograms, including both elastic & **elasto-plastic** deformations.
- 📖 Developed a **multi-phase field solver** to account for multiple phases in the lithiation model. Performed in depth analysis on multi-phase field **model parameters** and developed a **mapping** between the model parameters and physical factors of a real problem.
- 📖 Applied the **multi-phase field** model to the problem of **Peritectic** solidification. Demonstrated how the model parameters will be chosen for the given problem.
- 📖 Developed solvers using C++ based FEM library, **deal.II** & contributed a standalone code to its gallery.
- 📖 Enhanced code performance by 60% through **parallelization using PETSc** based wrappers in deal.II.

Research Experience (continued)

Data-driven modeling for predicting microstructural properties of Li-ion battery electrode.

Collaborators: Dr. Anand Agrawal & Dr. Narasimhan Swaminathan

- Solved the challenge of predicting the **electrode performance** of the Li-ion battery, based on **microstructural features** like grain size, in order to **optimize battery electrode design**.
- Generated voltammogram data via an **FEM** solver based on **deal.II libraries**, with Butler-Volmer boundary conditions and created microstructure meshes with **Microgen** and **GMSH**. Applied **Gaussian regression** to develop a nonlinear mapping between the voltammogram features and **grain size**.

Education

- 2019 – Present ■ **M.S.+Ph.D., Indian Institute of Technology Madras** in Mechanical Engineering.
CGPA: 9.39
- 2015 – 2019 ■ **Bachelors of Technology, Jamia Millia Islamia** in Mechanical Engineering.
CGPA: 9.66

Research Publications

Journal Articles

- 1 A. K. Agrawal, N. Swaminathan, and **U. Hussain**, “Probabilistic modelling of the nonlinear mapping between voltammograms and the grain size of the electrode material in li-ion batteries using optimally tuned gaussian process regression models,” *Journal of The Electrochemical Society*, vol. 172, no. 1, p. 013 501, 2025. 🔗 DOI: 10.1149/1945-7111/ada0b6.
- 2 **U. Hussain**, N. Swaminathan, and G. Phanikumar, “Numerical voltammetry of phase separating materials using phase field modeling,” *Journal of The Electrochemical Society*, vol. 171, no. 7, p. 070 502, Jul. 2024. 🔗 DOI: 10.1149/1945-7111/ad59cc.
- 3 **U. Hussain**, G. Phanikumar, and N. Swaminathan, “Mapping of multiphase field model parameters to physical factors in order to simulate desired phase transformations,” *Computational Materials Science*, vol. 226, p. 112 227, Jun. 2023, ISSN: 0927-0256. 🔗 DOI: 10.1016/J.COMMATSCI.2023.112227.

Conference Proceedings

- 1 **U. Hussain**, N. Swaminathan, and G. Phanikumar, “A multi-phase field framework application to the study of the peritectic reaction for an arbitrary material system,” in *Recent Advances in Mechanics of Functional Materials and Structures*, P. Kumari and S. K. Dwivedy, Eds., Singapore: Springer Nature Singapore, 2024, pp. 231–240, ISBN: 978-981-99-5919-8. 🔗 DOI: 10.1007/978-981-99-5919-8_20.

Open Source Contribution

- 1 **U. Hussain**, “Crystal growth using phase field modeling,” in *Code Gallery*, deal.II finite element library.
🔗 URL: https://dealii.org/developer/doxygen/deal.II/code_gallery_Crystal_Growth_Phase_Field_Model.html.

Conference Presentations

- 21-26 July 2024 ■ **16th World Congress on Computational Mechanics (WCCM 2024)**
Vancouver (Canada)

Conference Presentations (continued)

- 5-7 Sept 2023  **XVII International Conference on Computational Plasticity (Complas)**
Universitat Politècnica de Catalunya - BarcelonaTech (UPC), Barcelona (Spain)
- 11-14 Dec 2022  **8th Asian Conference on Mechanics of Functional Materials & Structures**
IIT Guwahati, Guwahati
- 29 June - 1 July 2022  **ME@75: Research Frontiers**
IISc Bangalore, Bengaluru

Teaching Experience

- July 2022 - Nov 2023  Doubt sessions for **Basics Of Finite Element Analysis – I & II, FEM: Variational Methods To Programming** at *NPTEL*.
- Feb 2022 - May 2022  Short course on **Computational Tools** at *PSG Institute of Technology*.

Projects

- Applied FEM  Solved the **SIR model** for disease spread in Uttar Pradesh using a self-developed **FEM code** in **Python** with **Picard iteration**, an **implicit time-marching** scheme, and a mesh from GMSH, achieving steady-state in 87 days.
- Engineering Plasticity  Implemented the **non-associative Drucker-Prager plasticity** model for pressure-sensitive materials in a self-developed **FEM code** in **MATLAB**, derived the tangent modulus, and validated its effectiveness against the J_2 model.

Miscellaneous Experience

Awards and Achievements

- 2019  Awarded with the prestigious **Prime Minister Research Fellowship (PMRF)**.
- 2017  **Scholarship awardee** during second year of B.Tech, earning a scholarship of 2500 INR.

Fellowships and Grants

- 2020-2024  Prime Minister Research Fellowship (**PMRF**) (Fellowship Amount: 3660000 INR, Research Contingency: 800000 INR).
- 2019-2020  Half-time Research Assistantship (**HTRA**) (Fellowship Amount: 372000 INR).

Workshops

- 2023  Workshop on **Technical and Scientific Writing**. *IIT Madras*.
- 2020  Faculty Development Program on **Advances in Theoretical and Computational Mechanics**. *NIT, Andhra Pradesh*.
- 2017  **Basics of Design and Innovation** Course. *Centre for Innovation and Entrepreneurship, Jamia Millia Islamia*.

References

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Dr. Gandham Phanikumar

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